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Project 2 Part 2: K-Means Strategy 2

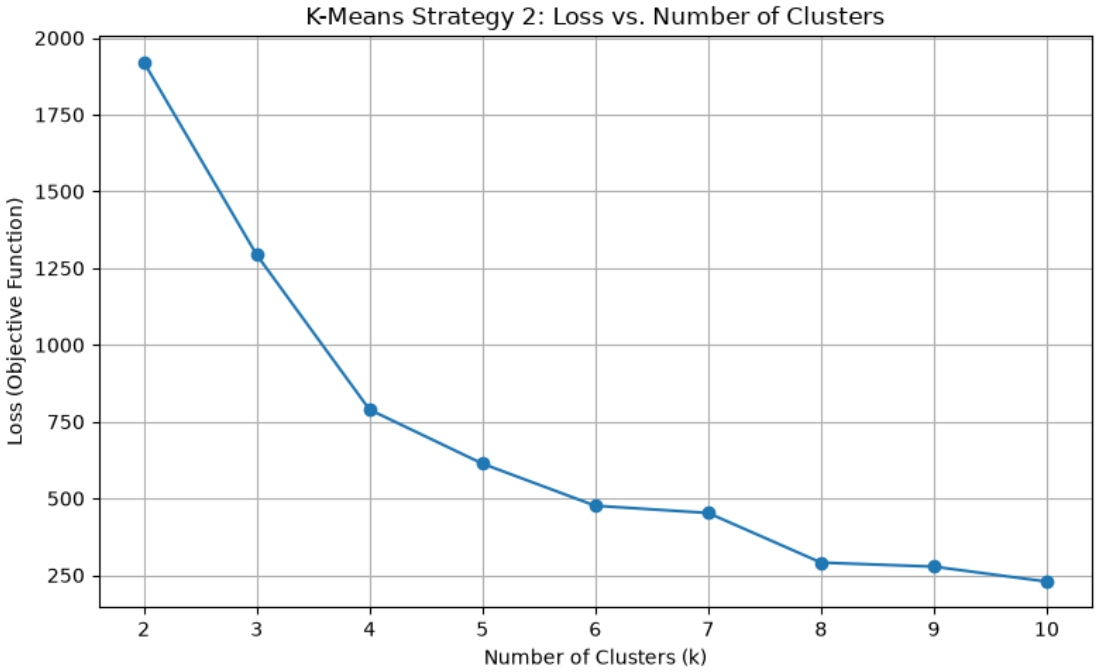
Project Analysis and Objective

The objective of Part 2 was to implement the K-means clustering algorithm using Strategy 2 and apply it to the provided dataset of 300 two-dimensional samples. The algorithm was evaluated for k = 2 through k = 10. For each value of k, the first initial centroid was obtained from the provided initial_S2() function, which used the last four digits of the student ID. Each remaining initial centroid was chosen as the sample whose average Euclidean distance to all previously selected centroids was maximal, while excluding samples already selected as centroids. After initialization, each sample was assigned to the nearest centroid using squared Euclidean distance. Each centroid was recalculated as the mean of the samples assigned to its cluster, and the assignment and update steps continued until the centroid coordinates were exactly unchanged between consecutive iterations. After convergence, the objective function was computed as $L = \Sigma(i=1 \text{ to } k) \Sigma(x \in Di) ||x - \mu i||^2$, where Di is cluster i and μi is its final centroid.

Solution Approach and Results

The same K-means procedure was applied independently for each value of k from 2 through 10 after generating the Strategy 2 initial centroids. Cluster memberships were recomputed at each iteration, and final assignments were determined using the converged centroids before evaluating the loss. The final objective-function values are shown in Figure 1.

Figure 1. K-Means Strategy 2: Loss vs. Number of Clusters



Observations and Analysis

The objective function decreased for every tested increase in k. The loss decreased from approximately 1921.03 at k = 2 to 229.05 at k = 10, an overall reduction of approximately 88.1%. One of the largest decreases occurred from k = 3 to k = 4 at approximately 39.0%, while

another substantial decrease occurred from k = 7 to k = 8 at approximately 35.7%. In contrast, the decreases from k = 6 to k = 7 and from k = 8 to k = 9 were only about 4.9% and 4.5%. The improvement from adding another cluster was therefore not uniform.

Compared with Strategy 1, Strategy 2 yielded a lower final objective value for seven of the nine tested values of k. At k = 2 and k = 8, its losses were approximately 23.1% and 20.9% lower than the corresponding Strategy 1 losses. However, Strategy 1 produced lower losses at k = 7 and k = 10. Therefore, Strategy 2 often improved the final solution, but it did not guarantee the lowest loss for every k because K-means can converge to different local solutions.

Figure 2. K-Means Strategy 2: Final Clustering for k = 10

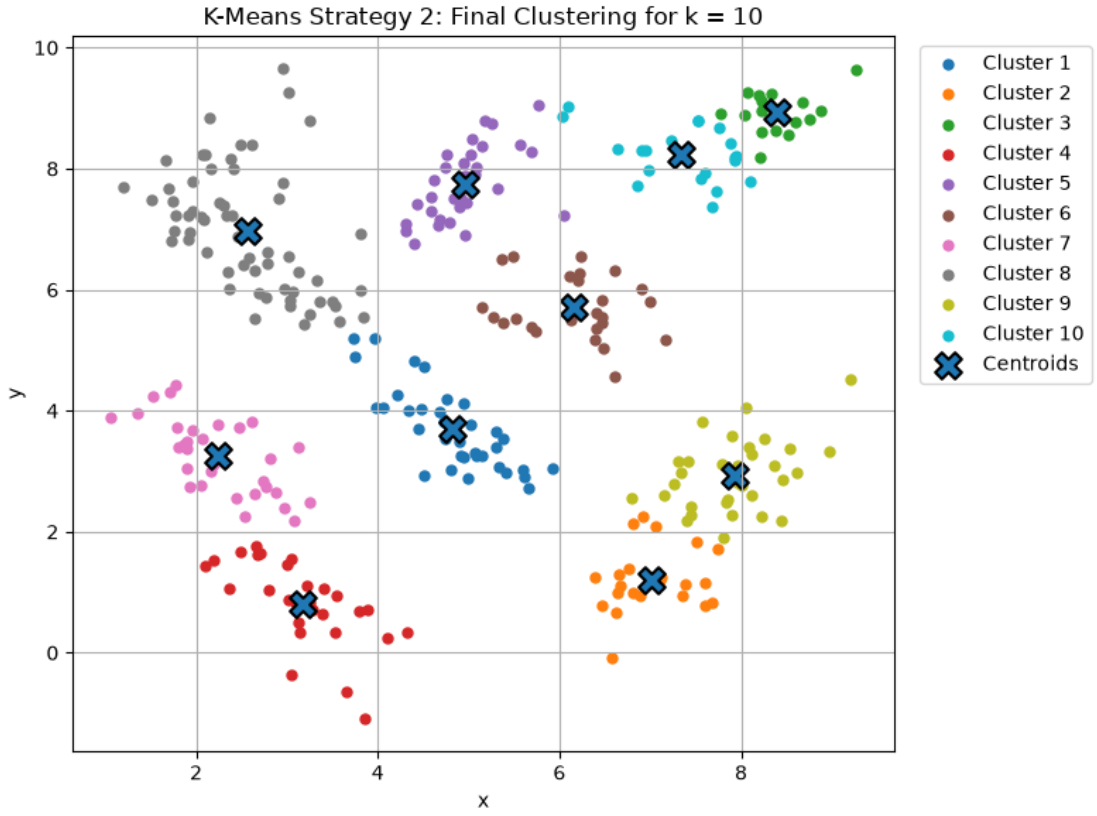


Figure 2 provides a geometric explanation for the decreasing objective function. At low values of k, each centroid must represent a broad region containing multiple groups of samples. As k increases, these regions are divided into smaller and more localized clusters. Consequently, the final centroids lie closer to their assigned samples, reducing the total within-cluster squared distance. At k = 10, the clusters closely follow the local groupings in the dataset, with each centroid positioned near the center of its assigned cluster. The completed implementation was verified by restarting the kernel and running the entire notebook from the beginning. It was then tested in Ed Lessons, where all 19 autograder test cases passed, yielding a score of 54/54 on the centroid and loss results. The results demonstrate that Strategy 2 can improve K-means clustering by spreading the initial centroids across the dataset, while the final solution remains dependent on initialization.